

# Stress Testing Ed-HMI220-070C

Nyle Controls Group

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## Abstract

In an effort to better detect behaviours of products at the limits of our operational range, we are stress testing devices at max loads for extended periods of time. **TLDR: HMI Max temp:87.1 °C (188.78 °F) while in a sealed and insulated box** The screen may behave like a heat sink, and orientation may impact cooling. at 100% load the HMI trends towards 61°C (140 °F)

## Method

The Ed-HMI220-070C in this testing sequence was subjugated to 100% CPU load for the duration of the workday 3.23.26 packed in an insulated box with 0 ventilation.

At the end of the day the Ed-HMI220-070C was removed from that box, and left running at 100% CPU load overnight.

## Tools

```
$ sudo apt install stress-ng lm-sensors
```

also get:

[https://github.com/elovejoy-nyle/loadTest/blob/main/cpu\\_burn.sh](https://github.com/elovejoy-nyle/loadTest/blob/main/cpu_burn.sh)  
which is a modified script from the authors of stress-ng package, with inclusions of logging.

lm-sensors was pre-configured on the distro which shipped with this device, if it were not, you would run:

```
sudo sensors-detect --auto
```

## Results

Nominal Run Time: 3 Days, in factory intended orientation, peak Temperature: 110F under not load HMI Max temp:87.1 °C (188.78 °F) while in an insulated, unventilated box.

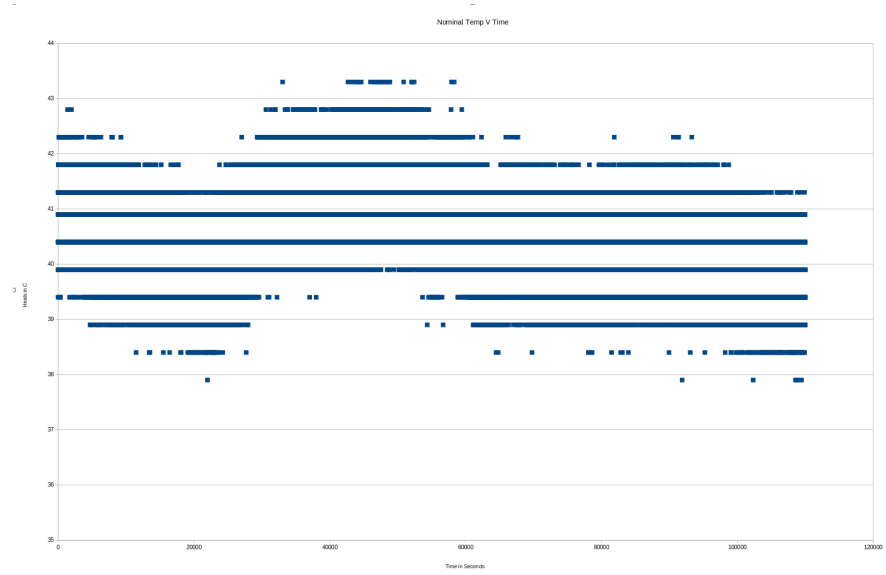


Figure 1: 1.25 Day, no load thermals

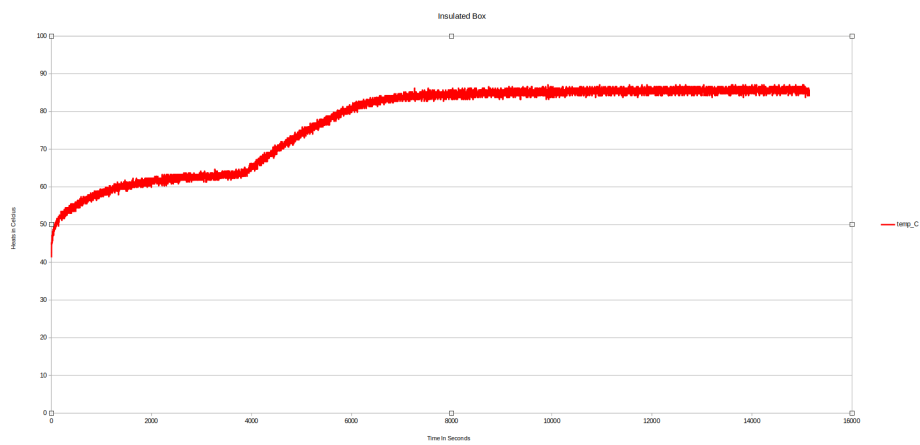


Figure 2: HMI in insulated box. Bump is where lid was closed

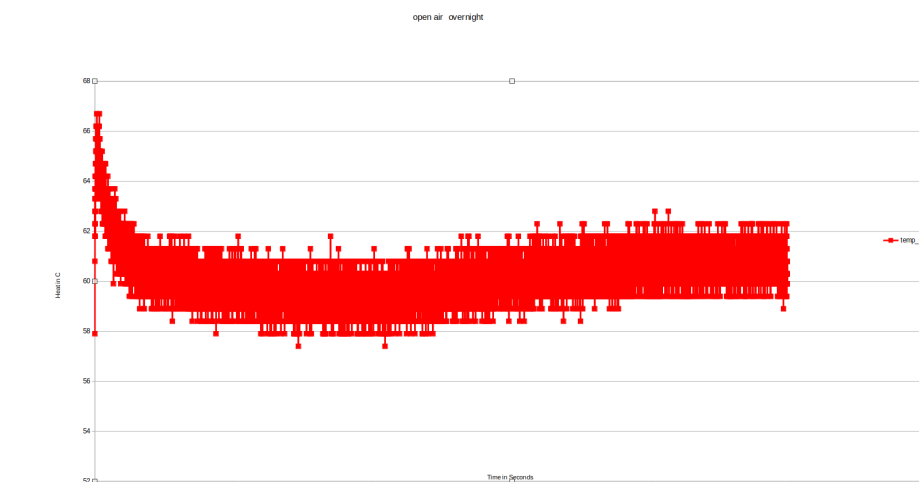


Figure 3: Test began immediately after insulated box testing

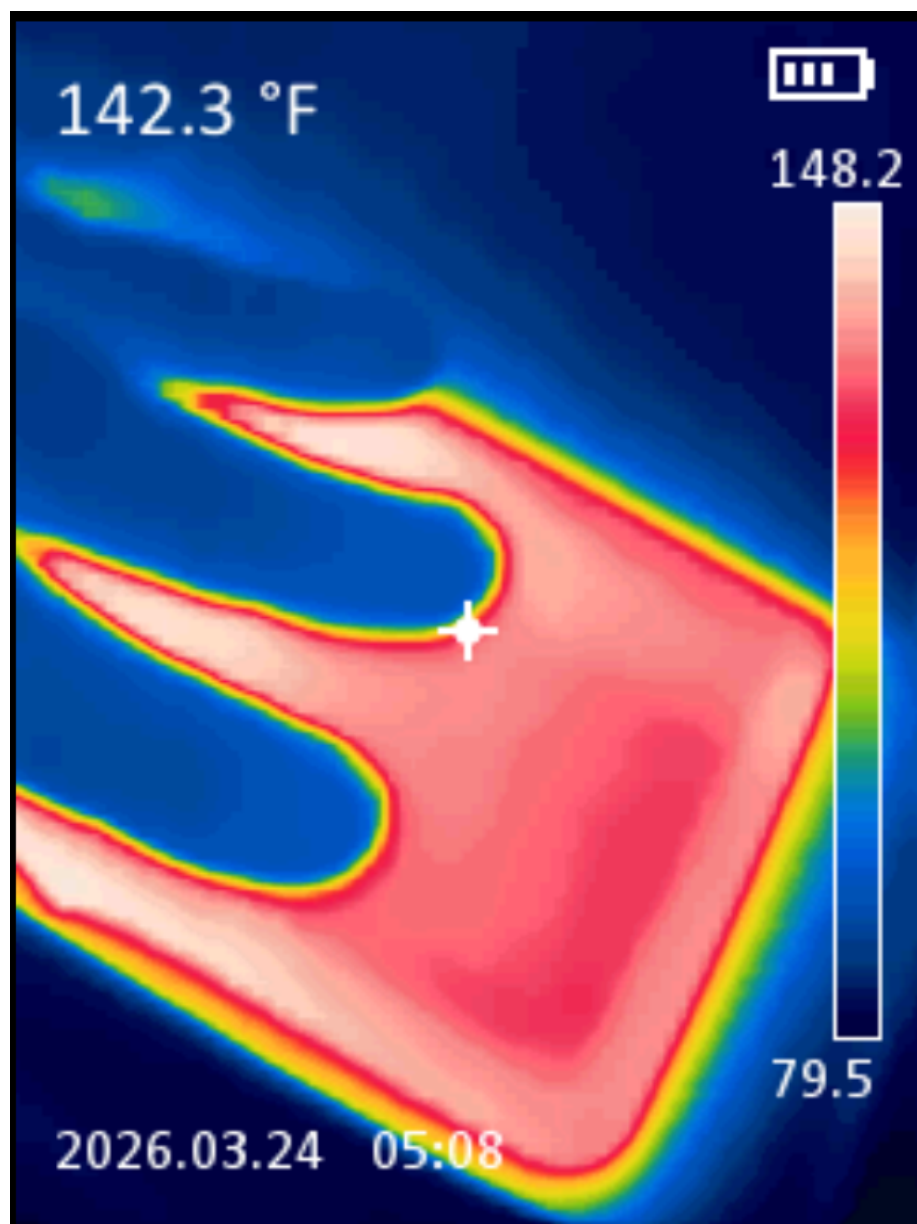


Figure 4: Human hand for contrast, after insulated test